

LunaSeis-1: When Balanced Apollo Seismic Detection Fails Under Cross-Station Continuous Scanning

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Abstract

Machine-learning models for lunar seismic detection are commonly developed from isolated event and background windows, but operational use requires low false-alarm rates across long continuous recordings and transfer to unseen stations. LunaSeis-1 constructs a checksum-verifiable Apollo Passive Seismic Experiment pipeline, reconciles 1,314 physical-event candidates, preserves Apollo timing/gap provenance, and evaluates lightweight detectors under event-, family-, chronology-, and station-aware controls. A 3,057-parameter tiny CNN failed two prospectively frozen continuous evaluations: it recovered 1/6 events with 2,932 false triggers over 2,591.47 station-hours, while an artifact-robust successor recovered 0/3 with 847 false triggers over 2,651.4 hours. A later 2,761-parameter depthwise CNN achieved 0.9115 mean development-event recall at 0.2106 merged triggers/hour. We froze an additional lower-confidence challenge containing 63 previously unexposed Grade-C natural impacts across 1,505.35 station-hours. The depthwise model recovered 12/63 at ± 180 seconds, produced 1,306 false triggers (0.868/hour), and retained 75.4% of the stream. These results do not establish an operational lunar detector. They demonstrate that balanced-window and development-continuous evaluation can substantially overstate Apollo cross-station utility, and provide a reproducible benchmark, failure analysis, and open research prototype.

1. Introduction

Apollo seismic recordings remain a uniquely important record of lunar interior activity. Compact automatic detectors could assist archive analysis and, in simulation, prioritize waveform retention. However, lunar signals are emergent, instrument characteristics differ across stations and years, catalog timing semantics are imperfect, and background recordings contain substantial acquisition artifacts.

LunaSeis-1 asks whether a lightweight raw-waveform model can generalize to an excluded Apollo station while maintaining useful continuous false-alarm behavior. The study prioritizes operational falsification over balanced-window accuracy. Its contribution is the combined reproducible data pipeline, strict leakage controls, continuous scanning, artifact diagnosis, and preservation of negative evidence.

This is not the first application of automated learning to Apollo seismic detection. Knapmeyer-Endrun and Hammer (2015) used hidden Markov models for Apollo 16 event detection and classification. Civilini et al. (2021) applied CNNs trained on a terrestrial station to Apollo PSE and LSPE data and motivated telemetry prioritization. Al-Qadasi and Bin Waheed (2026) subsequently evaluated compact Fourier Neural Operators on raw waveforms and spectrograms. Onodera (2024) also demonstrated that short-period archive analysis materially expands the known shallow-moonquake inventory. LunaSeis-1 therefore does not claim architectural priority. Its study focus is the combined cross-station, family-aware, continuous operational stress test and the transparent preservation of its negative outcome.

2. Data

Waveforms come from NASA PDS bundle `urn:nasa:pds:apollo_pse::1.0`. Labels originate from `urn:nasa:pds:apollo_seismic_event_catalog::1.0` and the corrected Onodera shallow-moonquake tables. The unified candidate registry contains 609 deep moonquakes, 623 natural impacts, 74 shallow moonquakes, and eight artificial impacts.

All downloaded products are checked against official sizes and NASA MD5 hashes. Raw products are not redistributed. MiniSEED nominal timestamps and ATT-derived timing remain separate. Gap sentinels are never silently interpolated.

3. Leakage controls

Physical events and known repeating families are indivisible within folds. Station-held-out models cannot use statistics from the test station. Background dates are selected independently of event dates. Architecture and threshold decisions use training/validation data only. Three continuous frames were selected and committed before their corresponding inference runs; each is permanently consumed.

The third frame was constructed only after the depthwise model and station thresholds were fixed. It contains 64 station-day observations representing 63 Grade-C natural impacts that were absent from all prior model splits and station-day manifests. Sixteen observations were selected per station by fixed SHA-256 rank from catalog records with positive MH visibility. All 3,426 previously exposed station-days were excluded, missing products could not be replaced, and waveform integrity was audited before inference. Grade C is weaker label evidence than Grades A/B, so this frame is reported separately as a lower-confidence challenge.

4. Models

Comparisons include energy, STA/LTA, handcrafted logistic regression, a 3,057-parameter tiny CNN, robust variants, a 2,761-parameter depthwise CNN, and a compact dilated TCN. Neural inputs contain robust-normalized waveform values and a separate validity mask.

5. Continuous-scanning protocol

Models scan 600-second windows at a 60-second stride. Candidate reference time is window start plus 120 seconds. Positive windows merge when adjacent inferred references are separated by at most 300 seconds. Primary one-to-one catalog matching uses ± 180 seconds, with ± 60 and ± 300 -second sensitivity analyses. False alarms are normalized by the union of locally integrity-qualified duration.

6. Results

6.1 First frozen test

Over 2,591.47 station-hours, the tiny CNN, logistic regression, and STA/LTA recovered 1/6, 1/6, and 0/6 eligible events. False-trigger rates were 1.131, 0.121, and 0.282 per hour; retained-duration fractions were 75.50%, 40.28%, and 97.92%.

6.2 Artifact diagnosis and second frozen test

Development data revealed strong station differences in quantized plateaus, steps, and extreme-value occupancy. Robust normalization reduced the tiny CNN's second-test false-trigger rate from 0.923 to 0.319/hour and retention from 62.48% to 45.58%. Nevertheless, robust CNN, original CNN, logistic, and STA/LTA all recovered 0/3 eligible events. Logistic regression remained best by false-trigger rate at 0.185/hour.

6.3 Later architecture development

At identical mean positive-development recall of 0.9115, the depthwise CNN produced 0.2106 merged triggers/hour, compared with 0.4102 for robust tiny CNN and 0.4558 for compact TCN. This selection occurred after both tests and cannot be promoted to untouched-test evidence. A full-day CLI smoke scan also revealed persistent activation.

6.4 Event-rich Grade-C challenge

All 256 planned products (97,596,992 bytes) passed exact-size and NASA-MD5 verification. All 64 selected event-station windows passed the frozen integrity gate, yielding 1,505.35 scannable station-hours. The selected observations represented 63 unique physical impacts because one event was independently selected at two stations. An additional 102 catalog references on the selected days were protected from false-trigger counting.

At the primary ± 180 -second tolerance, the frozen depthwise CNN recovered 12/63 physical events (recall 0.190), generated 1,306 false triggers (0.868/hour or 20.82/day), and retained 75.38% of the scannable duration. Precision excluding protected catalog triggers was 0.0091. Recall was 4/63 at ± 60 seconds and 16/63 at ± 300 seconds. The frame is now consumed and no threshold or model retuning is permitted.

7. Discussion

Balanced event/background windows did not predict continuous behavior. Station and acquisition regimes create powerful shortcuts, and trigger merging can hide nearly continuous positive-window activation behind apparently moderate trigger counts. Retained-duration reporting is therefore essential alongside trigger frequency.

The first two event denominators are small, but the third challenge provides a larger, deliberately weaker-label confirmation. Its 19.0% recall and 0.868 false triggers/hour reproduce the operational failure pattern despite the depthwise model's 91.15% development recall. The missed-event audit does not support a universal timing correction: score peaks occur widely over ± 2 hours and expanding tolerance mainly relabels background triggers.

8. Limitations

- The event-rich challenge uses Grade-C catalog labels, which are lower-confidence than the Grade-A/B development pool.
- Catalog references may not correspond to consistent physical arrivals.
- Many eligible test windows have weak raw signal evidence.
- Artifact proxy categories are not authoritative physical diagnoses.
- Calibration is not meaningful evidence when final-test recall is zero.
- The release model remains an experimental prototype.

9. Conclusion

LunaSeis-1 does not demonstrate an operational moonquake detector. Across 6,748.22 frozen station-hours, all tested neural variants failed the intended recall/false-alarm objective. The work demonstrates why apparently successful compact models can fail on continuous Apollo data and supplies a reproducible platform for improving them. Future confirmation requires better arrival-aligned labels, an explicit artifact taxonomy, and a new prospectively reserved high-confidence evaluation.

Data and code availability

Code, manifests, small checkpoints, configurations, predictions, and derived figures are prepared for public release. Raw NASA files are reconstructed from official PDS sources and are not mirrored. Release URLs and archival DOI will be inserted only after explicit publication authorization.

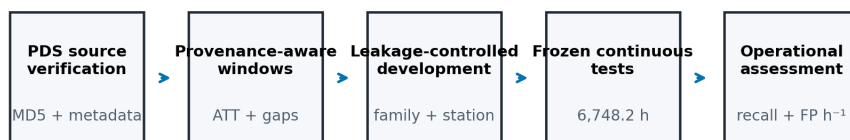
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Figures



Every test frame was selected and frozen before inference; consumed frames were never reused for tuning.

Figure 1. LunaSeis-1 study design. Product reconstruction, provenance-aware window construction, leakage-controlled development, and frozen continuous evaluation are separated explicitly. Consumed frames are never reused for tuning.

Frozen continuous evaluations remained below operational requirements

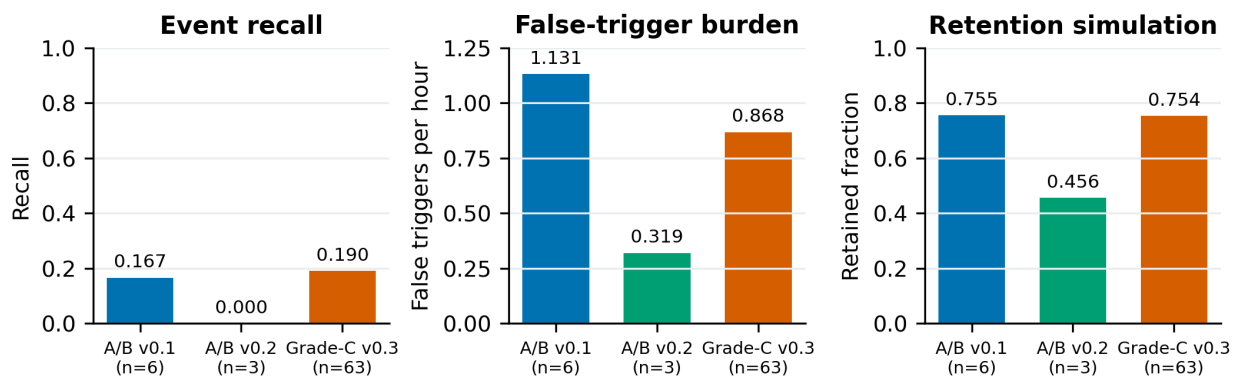


Figure 2. Primary ± 180 -second results for the three frozen continuous evaluations. Grade-A/B tests used the contemporary tiny/robust CNN; the lower-confidence Grade-C challenge used the later frozen depthwise CNN. Retained fraction measures the union of positive windows and exposes persistent activation not visible from merged-trigger counts alone.

Development selection substantially overstated operational transfer

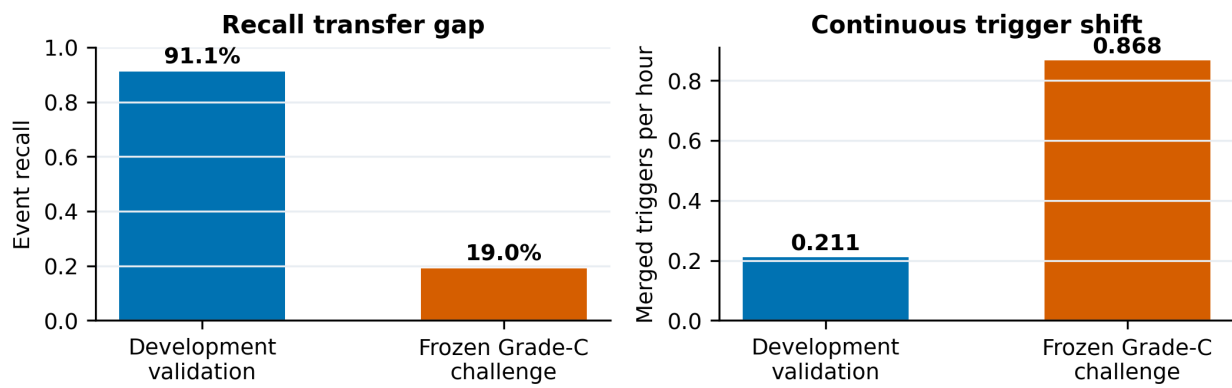


Figure 3. Transfer gap for the selected depthwise CNN. Development selection reported 91.15% mean event recall and 0.211 merged triggers/hour; the subsequent frozen Grade-C challenge yielded 19.05% recall and 0.868 false triggers/hour.

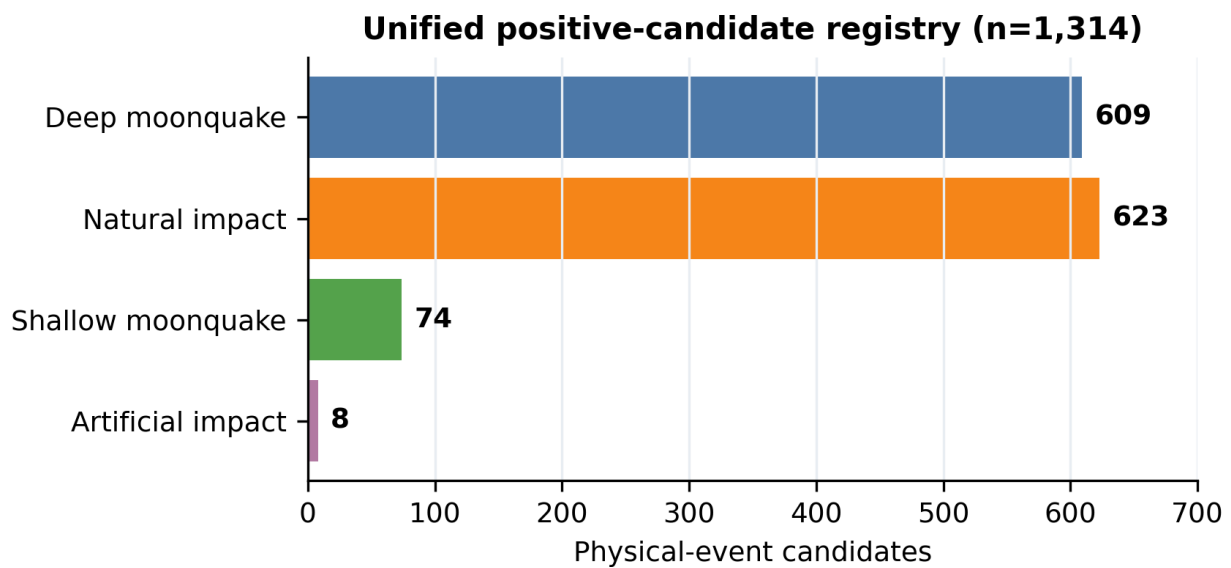


Figure 4. Physical-event composition of the unified 1,314-candidate registry. Counts are candidates with reconciled provenance, not automatically interchangeable training examples.